

AyurCentral HRMS - Technical Guide (for system owner / developer)

Audience: System owners and technical administrators. Complements the HR-facing manual.

Pair with: docs/AYURCENTRAL_HR_USER_MANUAL.md (share only that one with HR).

1. What the product is

Layer	Technology
UI	Single-page web app (HTML/JS) served by Google Apps Script web app (<code>doGet</code> → <code>Index.html</code> + lazy-loaded module HTML).
API	Apps Script server functions (<code>api*</code> entry points) called via <code>google.script.run</code> wrapper in <code>Scripts.html</code> .
Data	Google Spreadsheet tabs (Employees, PayrollRuns, PayrollInputs, etc.) via <code>DbService</code> .
Files	Google Drive (documents, payslip HTML, bulk upload staging).
Email	<code>MailApp</code> + <code>Notifications</code> sheet log.

Security model: UI hides routes by role, but **every API call re-checks** `PermissionService` / session. Spreadsheet editor access is **not** the app security boundary.

2. Repository layout (high level)

```
apps-script/src/
  foundation/    Auth, Db, Schema, Permissions, Constants, ApiFoundation
  ui/           Shell, Scripts.html, Styles, SettingsClient
  employee/     Directory, profiles, mandatory fields
  leave/       Types, balances, apply, approve
  payroll/     PayrollService, Compensation, Attendance bulk, Form T, Salary statement
  ats/         Jobs, candidates, hire letters, create employee from hire
  notifications/ Bell, inbox, email log
  tests/       Node-based string/regression tests (not full E2E)
  docs/        Integration notes + user manuals
```

Deploy: `apps-script/SETUP.md` - `clasp push`, `clasp deploy`, set web app URL.

Client cache bust: `HRMS.CLIENT_ASSETS_VERSION` in `Constants.gs` ; bootstrap returns `clientAssetsVersion` to the browser.

3. Authentication and users

- Sign-in: Google account → `AuthService` resolves session (`Users` sheet: `google_email` , `employee_id` , `role` , `status`).
- Roles: `OWNER` , `ADMIN` , `HR` , `MANAGER` , `EMPLOYEE` (see `03_USER_ROLES_PERMISSIONS.md`).
- OTP / session token stored client-side (`hrms_session_token`); expired session returns authorization errors on API calls.
- **OWNER** bypasses some module-matrix restrictions; treat as super-admin.

Navigation items: `PermissionService.NAV_ITEMS_` filtered by role; optional **Settings** → **HRMS module access by role** matrix persists overrides.

4. Routing and modules

`Scripts.html` maps **route** → **module id** for lazy load (`apiLoadModuleUi`):

Route	Module	Purpose
<code>payroll</code> , <code>payroll-run</code> , <code>salary-structure</code> , <code>compensation</code> , <code>attendance-bulk-upload</code> , <code>attendance-form-t</code>	payroll / payroll-attendance	Monthly payroll path
<code>salary-statement</code>	payroll-statement	CTC statement export
<code>employees</code> , <code>employee-profile</code> , <code>employee-mandatory-fields</code> , ...	employee	HR master data
<code>leave-*</code>	leave	Leave lifecycle
<code>ats-*</code>	ats	Recruitment
<code>notifications</code>	notifications	Inbox + email log
<code>settings</code>	settings	Admin configuration

The sidebar groups **Attendance** and **Payroll** as separate collapsible sections for HR users.

5. Payroll - backend design (what HR screens depend on)

5.1 Data flow

```

SalaryStructures (+ components) ← Compensation / salary-structure UI
    ↓
PayrollRuns (header: year, month, vertical_name, status)
PayrollInputs (per employee: days, daily_attendance_json, bonus, TDS, ...)
    ↓
PayrollService.calculate() → PayrollEngine
    ↓
PayrollRecords (snapshot + component_breakdown JSON)
    ↓
LOCK → PayslipService → Drive + Documents
  
```

5.2 Run IDs and verticals

Pattern	Example	vertical_name
Legacy combined	PR-2026-09	empty
Per vertical	PR-2026-09-SAPL	SAPL
Correction	PR-2026-09-SAPL-C1	same as source

- `findOpenRun_(year, month, vertical)` - only one open draft per vertical per month.
- `eligibleEmployees_` filters by employee `vertical_name` (or ID prefix) when run has a vertical.
- `AttendanceBulkService.assertRunVerticalMatches_` blocks uploading SAPL register into AOPL run.

5.3 Legacy import

- `importInputsFromLegacyRun / apiPayrollImportLegacyInputs` - copies inputs (including `daily_attendance_json`) from combined run to vertical run for matching employees.
- `tryAutoImportFromLegacy_` on `createRunInsideLock_` after `seedInputs_`.
- Client: `legacyImportHint`, `attendance_input_count` on run list, banners, `resolveTargetRunForPeriod_` fallback to legacy when vertical draft has zero attendance.

5.4 Workflow states

DRAFT → CALCULATED → UNDER_REVIEW → APPROVED → LOCKED

No unlock in Phase 1 - `createCorrectionRun` from LOCKED source seeds inputs from source (including `daily_attendance_json` on corrections).

5.5 Key APIs (Payroll)

API	Service method
<code>apiListPayrollRuns</code>	<code>listRuns</code>
<code>apiGetPayrollRun</code>	<code>getRunDetail</code> (+ sync eligible employees)
<code>apiCreatePayrollRun</code>	<code>createRun(year, month, vertical, notes)</code>
<code>apiCreatePayrollRunsAllVerticals</code>	<code>createRunsForAllVerticals</code>
<code>apiPayrollImportLegacyInputs</code>	<code>importInputsFromLegacyRun</code>
<code>apiSavePayrollInputs</code>	<code>saveInputs</code>
<code>apiCalculatePayroll</code>	<code>calculate</code>
<code>apiLockPayroll</code> / <code>apiFinalizePayroll</code>	lock / finalize
<code>apiCreatePayrollCorrection</code>	<code>createCorrectionRun</code>
Attendance bulk	<code>AttendanceBulkService.validateUpload</code> / <code>commitUpload</code>

Full list: `ApiPayroll.gs`.

6. Leave ↔ payroll bridge

- `PayrollLeaveBridge.getApprovedLopMapForPayroll` → `lop_from_leave` on inputs.
- `refreshLopFromLeave` / `applyLeaveLopToDays` - HR opt-in to copy into `lop_days` and recompute `paid_days`.
- Approved leave changes do **not** alter LOCKED runs.

7. Employee module (backend hooks)

- Create: allocate `employee_id`, seed `LeaveBalances`, optional Drive folder, audit log.
- `PayrollService.syncOpenPayrollRunsForEmployee` adds new active employees to open draft runs.
- Mandatory fields: configurable required columns for create/bulk (employee-mandatory-fields route).

8. ATS → HRMS hire path

- Stages in `AtsConstants` / `AtsEngine`; terminal hire stage `HIRED`.
- Letters: PDF generation + email (templates in Settings + code-managed HTML layouts).
- `CompensationService` + `PayrollEngine` for offer salary breakup on PDF.
- Create employee from hire: RPC creates `Employees` row with new ID; links application to employee.

See `docs/ATS_INTEGRATION_NOTES.md`.

9. Notifications

- Event types: leave submitted/approved/rejected, payroll review, payslip available, etc.
- Rows in `Notifications` sheet; HR views delivery log in app.
- Toggles in Settings → email section.

See `docs/NOTIFICATIONS_INTEGRATION_NOTES.md`.

10. Settings and schema maintenance

- **Settings** sheet: booleans, role module matrix, content templates, company emails.
 - **Database setup:** HRMS menu in spreadsheet or `SchemaService.ensureSheetHeaders` - adds columns such as `PayrollRuns.vertical_name`, template rows.
 - If Settings templates missing, UI shows warning to run database setup.
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11. Deploy and verify

```
cd apps-script
clasp push
clasp deploy # or update existing deployment
```

Pull the latest application source from your repository before deploying, if your team uses version control.

Verify in browser:

1. Footer on payroll pages shows `clientAssetsVersion` (version string updates when IT deploys a new build).
2. Sidebar shows Attendance + Payroll sections for HR roles.
3. Combined legacy + import banner on vertical runs when older combined data exists.

Tests (local):

```
node tests/payroll-vertical-runs.test.js
```

12. Operational troubleshooting

Symptom	Likely cause	Action
HR sees old UI	Deployment not updated or cached	New deployment + hard refresh; check <code>CLIENT_ASSETS_VERSION</code> .
Empty vertical payroll	Data on <code>PR-YYYY-MM</code>	Import legacy or open combined run.
Calculate skips employees	<code>MISSING_STRUCTURE</code> / zero working days	Fix structure or attendance.
Cannot finalize	Still DRAFT or exception flags	Calculate; resolve Payroll check blockers.
403 on API	Role or module matrix	Users sheet role; Settings module access.
Ask HR errors	Feature flag / API key	Settings + <code>AskHr</code> service config.

13. Spec documents (source of truth)

Doc	Topic
<code>00_MASTER_SPEC.md</code>	Product scope
<code>03_USER_ROLES_PERMISSIONS.md</code>	Roles matrix
<code>04_EMPLOYEE_MODULE.md</code>	Employee master
<code>05_LEAVE_MODULE.md</code>	Leave workflow
<code>06_PAYROLL_COMPENSATION_MODULE.md</code>	Payroll math and states
<code>07_DASHBOARD_NOTIFICATIONS.md</code>	Dashboard + mail events
<code>09_APPS_SCRIPT_ARCHITECTURE.md</code>	Code structure

14. Regenerating the PDF manuals

From the `docs` folder:

```
python3 build-ayurcentral-manuals-pdf.py
```

This writes `AYURCENTRAL_HR_USER_MANUAL.pdf` and `AYURCENTRAL_TECHNICAL_GUIDE_FOR_ADMIN.pdf` next to the markdown sources.

Internal technical guide - not for distribution to end-user HR staff.